

Phat Nguyen Thuan

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Education

University of Information Technology – VNUHCM

Bachelor of Science in Artificial Intelligence

Cumulative GPA: 8.59/10.0 (through 6th semester)

2023 – 2027

HCMC, Vietnam

Skills

- **LLM Inference & Systems Optimization:** Research into PagedAttention, FlashAttention/FlashAttention-3, continuous/inflight batching, KV-cache management, speculative decoding, quantization, and throughput/latency trade-offs; hands-on with vLLM and llama.cpp.
- **Multimodal AI & Computer Vision:** Vision-language and cross-modal retrieval with CLIP and BEiT-3; 3D object retrieval, video understanding, object detection, transformer architectures, Weighted Box Fusion, and large-model evaluation.
- **Programming, Acceleration & Infrastructure:** Python, PyTorch, JavaScript/TypeScript; GPU benchmarking and model training/inference on NVIDIA GH200 and A100; ARM64 porting, xFormers/FlashAttention builds, FastAPI, Node.js, Docker, and Linux.
- **Agentic AI, MCP & Evaluation:** MCP servers and tool integration; multi-agent orchestration, RAG, agent workflows, LLM-as-evaluator experiments, and reproducible benchmarking on LoCoMo and LongMemEval.
- **Frameworks:** LangGraph, CrewAI, LangChain, and LlamaIndex; Node.js/Express, SQLAlchemy/Prisma.

Projects & Research

- **ByteRover: Agent-Native Memory Through LLM-Curated Hierarchical Context** [\[arXiv\]](#)
Co-author, preprint (2026). LLM-curated hierarchical memory with progressive retrieval and adaptive lifecycle; state-of-the-art on LoCoMo and competitive on LongMemEval.
- **SHREC 2025: Retrieval of Optimal Objects for Multi-modal Enhanced Language and Spatial Assistance (ROOMELSA)** [\[DOI\]](#)
Co-author, Computers & Graphics 132 (2025); 1st place in ROOMELSA. Language- and spatially-guided 3D object retrieval.
- **A Hybrid Video Retrieval System Using CLIP and BEiT-3 for Enhanced Object and Contextual Understanding** [\[DOI\]](#)
Co-author, Springer CCIS (2025); 2nd place in HCMC AI Challenge 2024. CLIP/BEiT-3 visual-textual video event retrieval.
- **Robust Traffic Vehicle Detection under Real-World Conditions with Misclassified Vehicles Minimization and Weighted Box Fusion** [\[IEEE\]](#)
Co-author, IEEE MAPR 2025; 2nd place in SoICT Hackathon 2024. Robust traffic detection with Co-DETR, D-FINE, misclassification correction, and model fusion.

Professional Experience

Amaris Consulting

07/2026 – Present

Junior AI Specialist

- Currently developing an **MCP server for AMAIA Services**, unifying platform capabilities behind an interface that coding agents such as Codex and GitHub Copilot can consume.
- Run research and evaluations of MCP-enabled agent workflows for AMAIA's agentic software-testing platform, assessing tool use, reliability, and integration patterns for software quality assurance.

ByteRover

07/2025 – 06/2026

AI Engineer & Developer Advocate

Remote

- Led evaluation work for ByteRover's agentic memory system, improving LoCoMo and LongMemEval results and contributing to state-of-the-art LoCoMo accuracy [\[benchmark\]](#).
- Co-authored the team paper on agent-native hierarchical memory architecture, covering the Context Tree, LLM-curated memory, progressive retrieval, and adaptive knowledge lifecycle [\[arXiv\]](#).
- Contributed to evaluation pipelines and the open-source ByteRover CLI agentic framework [\[GitHub\]](#).

Talks

- **OpenInfra & Cloud Native Day Vietnam 2025** — *Benchmarking LLM Performance and Efficiency Across GH200, H100, and A100.*
- **HCMC AI Meetup #1, Menlo Research** — *Training on NVIDIA GH200: Optimizing for ARM64 Architecture.*

Certifications & Awards

- **Overall Band 8.5** – IELTS Exam March 2024
- **2nd Prize** – TRAE SOLO Hackathon: Unbound Creativity (Video Generation Track) May 2026
- **2nd Prize** – HCMC AI Challenge 2024 (Video Event Retrieval) Dec 2024
- **2nd Prize** – BKAI-SoICT Hackathon 2024, NAVER Track Nov 2024